

Day = 14

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Q17 Can Atal Tinkering Lab Transform

STEM education in India? Analyze
potential impact on Scientific Temperament
and innovation.

Ans Recently Govt has proposed 50,000 ATL
in school to promote Scientific temperament
and Curiosity. Tinkering Labs give approach
to implement theory in practical way.

STEM EDUCATION

↓ Science ↓ Technology ↓ Engineering ↓ Maths

CAN Atal Tinkering Lab Transform?

Feasibility

1) Foster Critical
thinking and problem
solving

Challenges

1) Lack of Skilled
Trained Teacher.
in STEM.

2) explore STEM Concept
through Hand on
activities

2) 105 Lakh Grant
school have no Science
Teacher
↓
(Ashoka + Princeton)
University research

Potential Impact of ATL :->

- 1) promote logical reasoning, Creative thinking
eg NITI Aayog = Tinkering in School report
↓
30% rise in logical reasoning
- 2) Next Generation of innovator and Tech Leader
- 3) Industry Academia Collaboration increased

Provision ATL Labs in Budget aim to promote
Scientific Temperament. By increasing
skilled & STEM teacher, it can promote
AI Driven future and make Global
Leader in STEM education

Q2 How Can National Centre of Excellence for skilling bridge India's skill gap?
Critical analyze structural and technological challenges they face.

Ans National skilling programme aim to enforce industry linked skilling programme that equip youth with skill required for manufacturing.

Ways To bridge Skill Gap :-

- i) Partnered with Global Companies
 - NCE partnered to offer Training that required
- ii) Reducing gap between Classroom learning and industrial demand
eg: Quantum Computing, Big data Analytics, Robotics Lesson

III) Promote Sustainable Technology

eg Cyber security, Cloud Computing

WEF report = future of 2023 = 8.5 Gross job till 2030 in digital sector

Challenges

Technological ⇒

Potential

• 8.5 Gross job in digital till 2030

• Expansion of Solar ~~and~~ sector in India

Challenges

• Lacking technology to/skill to access
↓

Technological Unemployment

• Imported component from China called

"missed opportunity"

Structural
Unemployment

• Abundance of human resource Capital

• Industry Academia Gap

Mckenzie report

5 Million job vacant in last 10 due to skill gap

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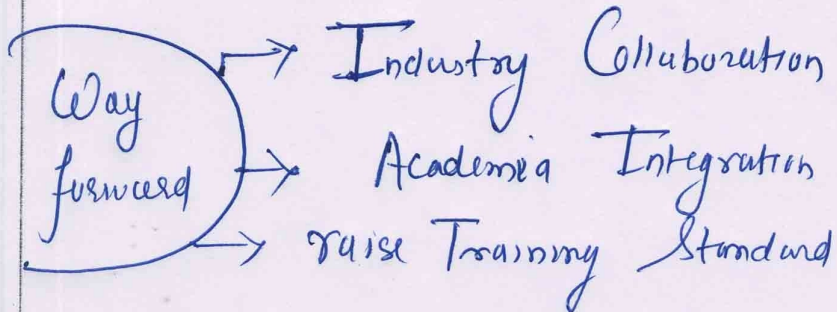
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India has a huge demographic dividend
However challenges like Skill gap, and
Technology ~~up grad~~ unemployment need to
address by Global Collaboration which
steer to make Leader in Technology

Q3 Discuss role of education and skill development in sustaining India's long term economic growth. How can strategic investment in human capital bridge gap between economic aspiration and realities.

Ans Education and Skill development are major tools which helps to obtain benefit of huge demographic dividend.

Education in native language increase learning ability of individual
In Noam Chomsky book "Language & Mind"
highlight cognitive ability enhanced in
Native language.

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Role of education and Skill development in economy growth

1) Education and Learning Concept

- STEM education increased Creative thinking
- Experiential Learning by practical approach
- Cognitive Development, Analytical Thinking

2) Technology Integration in Education

- Use of AI, Machine Learning, IoT, Big Data Analytics

3) Skilling and Workforce Development

- Industry Linked Skilling program
- Train the Trainer Program
- Public Private Partnership model to develop Skilling

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<u>Economic Aspiration</u>	<u>Reality</u>
1) <u>Inclusion development</u> ↓ equitable distribution of economic benefit	2) <u>Income Desparity</u> Top 1% Concentrate 40% of total wealth & Bottom 50% shared 3% only (Oxfam report)
2) <u>High income status</u> ↓ aims to transit to high income economy	1) India fall in middle income trap
3) <u>Manufacturing hub</u>	11) 10 year of Make In India Still 27% GVA
4) <u>Skilled workforce</u>	10) only 4.7% got formal skill training
5) <u>Fintech economy</u>	11) <u>RBI Report</u> ↓ 72% farmer still rely on Cash to sell

Strategies to bridge gap between aspiration and reality →

① | Industry Linked Skilling Program

- reduce discrepancy between skill exist and required.

2) | AI Accessibility in Rural education

3) | Industry Academia Partnership ↓ for practical Training & research

4) | Human Capital Development ↓ eg education, Health, to increase efficiency

5) | Sustainable Economic Growth ↓ Balancing inclusive and innovative development

6) | Cutting edge Tech, Research & Development ↓ eg AI, robotics etc

Challenges and Solutions

<u>Challenges</u>	<u>Solution</u>
i) Resistance from education perceived AI as Threat	i) Training Module that position AI as Enabler
ii) Insufficient funding ² research	ii) joint research initiatives with global institution
iii) Industry-Academia gap. eg 60% engineering graduate Lack skill	iii) Train the Trainer program.

Investment in education and skill development is Strategic necessity for economic future to harness opportunity.

In David book "Work for future" tells most employment will created in digital sector. There is need of upgrade skill to utilize opportunities

Q4 The Integration of Artificial intelligence in India education system has potential to revolutionize Learning. Critically assess feasibility of AI driven educational reform in India, highlighting ethical, infrastructural and pedagogical concerns.

Ans AI driven education has Capability to transform India's Learning Ecosystem. It provides personalised Learning experience which enhance accessibility and quality of education.

Stanford
University
report

AI Index =
Report 2023

AI enabled education
improve Student
Learning by 25 %.

Role of AI in Indian Education

- i) Make India global Leader in
STEM education
eg Science, Technology, Engineering, Math
- ii) Build Scientific Temper, Innovation
and Creative thinking
- iii) Enhanced Access of education in rural area
- iv) Increased Learning outcome and
Cognitive Skill
eg 25% increase Learning in AI Index report
by Stanford
- v) Potential to make Hub for Skilling

Feasibility of AI Driven Education

Concern → feasibility

a) Ethical : i) AI Accessibility
in Rural education
to bridge digital divide

ii) AI Transparency
and Accountability
↓
Build Trust in AI

iii) Responsible AI
Deployment

b) Infrastructure

i) Industry - Academia
Partnership for
practical Training

Challenges

i) Lacking access
of infrastructure of AI
in rural

ii) Issue of Data
privacy

iii) Unethical use AI
in privacy, hacking etc.
voice & cloning

i) AICTE report
60% graduate unemployable
due to lacking skill

i) Sector Skill Council that define skilling requirement

ii) Shortage of Trainers proficient in AI, data Science.

iii) Adoption of Global Skill framework

iv) Technical education non aligned with International Standard.

v) Pedagogical

i) AI Driven Assessment System for student evaluation

i) Teacher Lacking utility of AI in assessment.

ii) AI driven education platform

ii) Lack of High quality Technical Content



III) STEM and AI
integrated education system

III) Shortage of STEM
Trained teacher.

IV) Digital Learning
Platform for adaptive
use of AI

IV) Lack of Teacher Training
in AI, Machine Learning
IoT.

WAY FORWARD

- I) Global partnership for joint Training
- II) Low cost AI platform for adoption in rural
- III) Training Module for AI as a Enabler.

Integration of AI in education can ^{enhance} ~~transform~~
learning system. Given in Stanford University

AI report, Learning increased by 25%. A

proper framework is required to address
challenges, and bring digital revolution in
education sector